THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If light enters a medium of refractive index 1.5 from vacuum, its speed becomes ($c = 3 \times 10^8$ m/s) 1

(a) 2×10^8 m/s **(b)** 4.5×10^8 m/s **(c)** 3×10^8 m/s **(d)** 1.5×10^8 m/s

2. The power of a concave lens is 1

(a) negative **(b)** positive **(c)** infinite **(d)** zero

3. A concave mirror forms a real, inverted and same-sized image when the object is placed 1

(a) at the centre of curvature **(b)** at infinity
(c) at the focus **(d)** between pole and focus

4. Snell's law states that for two media the ratio of sin i to sin r is 1

(a) equal to tan i **(b)** zero
(c) equal to 1 always **(d)** constant

5. A convex lens gives a real, inverted, same-sized image when the object is placed 1

(a) at F **(b)** at 2F **(c)** at infinity **(d)** between F and O

6. A concave mirror forms a virtual, erect and magnified image when the object is placed 1

(a) at C **(b)** beyond C
(c) between the pole and the focus **(d)** at the focus

7. A rectangular glass slab produces 1

(a) no change at all **(b)** lateral displacement of the emergent ray
(c) dispersion into 7 colours **(d)** deviation of the ray by 90 degrees

8. A convex lens used as a magnifying glass has the object 1

(a) at 2F **(b)** between optical centre and F
(c) beyond 2F **(d)** at infinity

9. Which mirror is used by a dentist to see an enlarged image of teeth? 1

(a) Concave mirror **(b)** Plane mirror **(c)** Convex mirror **(d)** Any of these

10. Which medium has the highest optical density? 1

(a) Air **(b)** Glass **(c)** Water **(d)** Diamond

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Define 1 dioptre of power. 2

12. What is the nature of the image formed by a concave lens? 2

13. Name the type of mirror used in 2
(a) headlights
(b) rear-view mirrors.

14. State Snell's law. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A concave mirror has a radius of curvature of 40 cm. Find its focal length and the image position of an object at infinity. 3

16. Define power of a lens. A lens has focal length -50 cm. Find its power. 3

17. Distinguish between real and virtual images with one example each. 3

OR

Draw ray-diagram reasoning: describe the image formed by a concave mirror when the object is placed between the focus and the pole. 3

18. State three uses of concave mirrors. 3

OR

Explain the phenomenon of the apparent raising of a coin in a vessel of water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Define refraction. State the laws of refraction. Calculate the refractive index of a medium in which the speed of light is 2×10^8 m/s. 5

OR

A concave mirror of focal length 15 cm forms an erect image of an object placed 10 cm from it. Find the image distance and magnification. 5

20. Trace the path of a ray of light through a rectangular glass slab. Explain why the emergent ray is parallel to the incident ray. 5

OR

Find the refractive index of a substance if light enters it from air at an angle of incidence 60 degrees and the angle of refraction is 30 degrees. Also find the speed of light in it. 5

END OF QUESTION PAPER — Total: 40 Marks

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